TM Forum Specification

Entity Inventory API REST Specification

TMF703

Team Approved Date: 15-Sep-2021

Release Status: Production	**Approval Status: TM Forum Approved**
Version 4.0.0	**IPR Mode: RAND**

©TM Forum 2021. All Rights Reserved.

Notice

Copyright © TM Forum 2021. All Rights Reserved.

This document and translations of it may be copied and furnished to others, and derivative works that comment on or otherwise explain it or assist in its implementation may be prepared, copied, published, and distributed, in whole or in part, without restriction of any kind, provided that the above copyright notice and this section are included on all such copies and derivative works. However, this document itself may not be modified in any way, including by removing the copyright notice or references to TM FORUM, except as needed for the purpose of developing any document or deliverable produced by a TM FORUM Collaboration Project Team (in which case the rules applicable to copyrights, as set forth in the TM FORUM IPR Policy, must be followed) or as required to translate it into languages other than English.

The limited permissions granted above are perpetual and will not be revoked by TM FORUM or its successors or assigns.

This document and the information contained herein is provided on an "AS IS" basis and TM FORUM DISCLAIMS ALL WARRANTIES, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTY THAT THE USE OF THE INFORMATION HEREIN WILL NOT INFRINGE ANY OWNERSHIP RIGHTS OR ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

TM FORUM invites any TM FORUM Member or any other party that believes it has patent claims that would necessarily be infringed by implementations of this TM Forum Standards Final Deliverable, to notify the TM FORUM Team Administrator and provide an indication of its willingness to grant patent licenses to such patent claims in a manner consistent with the IPR Mode of the TM FORUM Collaboration Project Team that produced this deliverable.

The TM FORUM invites any party to contact the TM FORUM Team Administrator if it is aware of a claim of ownership of any patent claims that would necessarily be infringed by implementations of this TM FORUM Standards Final Deliverable by a patent holder that is not willing to provide a license to such patent claims in a manner consistent with the IPR Mode of the TM FORUM Collaboration Project Team that produced this TM FORUM Standards Final Deliverable. TM FORUM may include such claims on its website but disclaims any obligation to do so.

TM FORUM takes no position regarding the validity or scope of any intellectual property or other rights that might be claimed to pertain to the implementation or use of the technology described in this TM FORUM Standards Final Deliverable or the extent to which any license under such rights might or might not be available; neither does it represent that it has made any effort to identify any such rights. Information on TM FORUM's procedures with respect to rights in any document or deliverable produced by a TM FORUM Collaboration Project Team can be found on the TM FORUM website. Copies of claims of rights made available for publication and any assurances of licenses to be made available, or the result of an attempt made to obtain a general license or permission for the use of such proprietary rights by implementers or users of this TM FORUM Standards Final Deliverable, can be obtained from the TM FORUM Team Administrator. TM FORUM makes no representation that any information or list of intellectual property rights will at any time be complete, or that any claims in such list are, in fact, Essential Claims.

Direct inquiries to the TM Forum office:

181 New Road, Suite 304
Parsippany, NJ 07054 USA
Tel No. +1 862 227 1648
TM Forum Web Page: www.tmforum.org

© TM Forum 2021. All Rights Reserved.

Table of Contents

Notice 2
Table of Contents 4
List of Tables 6
Introduction 7
SAMPLE USE CASES 8
Entity Inventory Use Cases 8
Support of polymorphism and extension patterns 9
RESOURCE MODEL 10
Managed Entity and Task Resource Models 10
Entity resource 10
Managed Entity resource 11
Association resource 21
Notification Resource Models 26
Entity Create Event 28
Entity Change Event 28
Entity Delete Event 28
Association Create Event 29
Association Change Event 29
Association Delete Event 29
API OPERATIONS 31
Operations on Entity 31
List entities 31
Retrieve entity 40
Create entity 45
Patch entity 54
Delete entity 59
Operations on Managed Entity 60
Operations on Association 60
List associations 60
Retrieve association 69

© TM Forum 2021. All Rights Reserved.

Create association 71
Patch association 75
Delete association 78
API NOTIFICATIONS 79
Register listener 79
Unregister listener 80
Publish Event to listener 80
ROLE BASED ACCESS CONTROL 82
Acknowledgements 83
Version History 83
Release History 83
CONTRIBUTORS TO DOCUMENT 84

© TM Forum 2021. All Rights Reserved.

List of Tables

N/A

© TM Forum 2021. All Rights Reserved.

Introduction

The Entity Inventory API allows the management of the entire lifecycle of the inventory elements including (managed entities and associations). Entity Inventory is primarily intended to address those entities which may not be addressed by any of Product, Service or Resource inventories. However, with an extension pattern, it can be used to address such inventory requirements as well.

The Entity Inventory represents two base elements: entity and association. By extending these elements you may make a concrete model of instance of any 'Thing'. So, the entity inventory can be used in any of Internet of Things (IoT) use cases.

This API also provides ManagedEntity Resource as an extension of base entity that could be future extended for different applications. There are examples of such a managed entity provided in this document.

© TM Forum 2021. All Rights Reserved.

SAMPLE USE CASES

Entity Inventory Use Cases

The entity inventory API allows the lookup of the inventory elements to support relevant business processes such as:

UC1: A query running on the inventory entities corresponding to a given related party, for example a customer

UC2: An inventory for all (physical/logical) assets belong to a supplier with CRUD operations.

UC3: An image of virtual network deployed function instances as its TOSCA templates onboarded into an entity catalog.

The examples provided in this document demonstrate how to support the above use cases.

© TM Forum 2021. All Rights Reserved.

Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance Graph and ToscaNode inheriting properties from the abstract Entity object.

Generic support of polymorphism and pattern extensions is described in the TMF API Design Guidelines v4.0 (TMF630) document.

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of ManagedEntity instances some may be instances of ToscaNode where other could be instances of Graph. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance an EntitySpecificationRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as EntitySpecificationRef, and not the class type of the referred object.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected *characteristics* of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended. For example, a ToscaNode Resource could be an extension of a ManagedEntity.

© TM Forum 2021. All Rights Reserved.

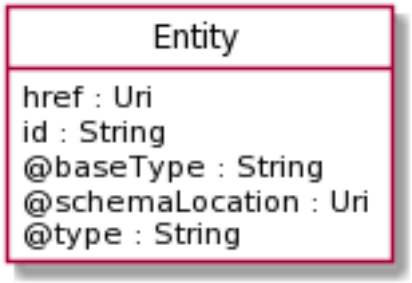
RESOURCE MODEL

Managed Entity and Task Resource Models

Entity resource

Base entity schema for use in TMForum Open-APIs.

Resource model



Field descriptions

Entity fields

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

Json representation sample

We provide below the json representation of an example of a 'Entity' resource object

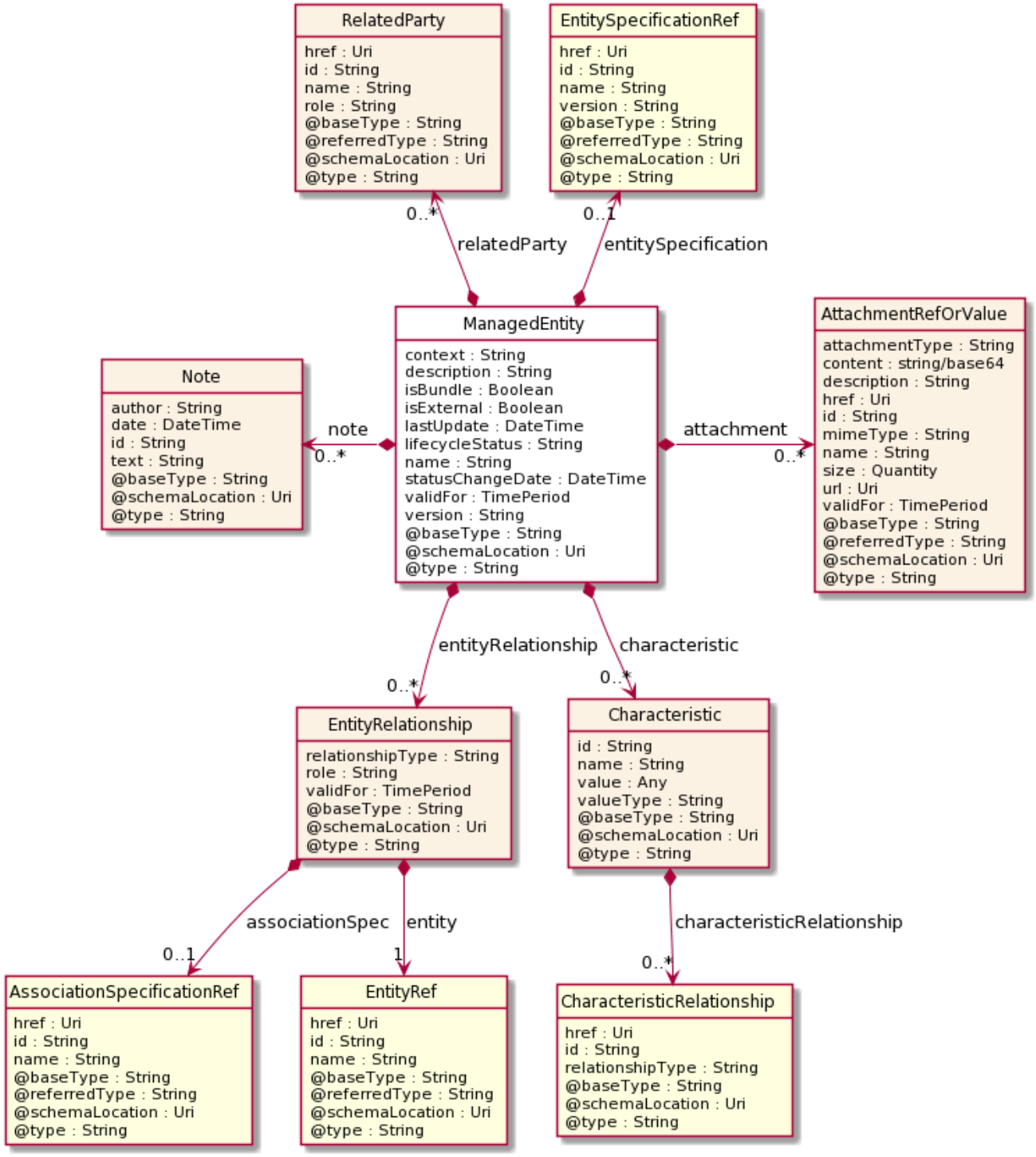

```
{
   "id": "8999",
   "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8999",
   "@type": "Entity",
   "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Entity.schema.json",
   "@baseType": ""
}
```


© TM Forum 2021. All Rights Reserved.

Managed Entity resource

ManagedEntity represents an entity that describes the common set of attributes shared by all concrete entities (e.g. Resource, Service, ..) in the Entity Inventory. A ManagedEntity may concern a simple concept, such as a logical point of communication, or a complex concept, such as an Application or System .

Resource model



© TM Forum 2021. All Rights Reserved.

Field descriptions

ManagedEntity fields

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
attachment	A list of attachment ref or values (AttachmentRefOrValue [*]). Attachments that may be of relevance to this entity, such as picture, document, media.
characteristic	A list of characteristics (Characteristic [*]). Describes a given characteristic of an object or entity through a name/value pair.
context	A string. A string used to give a context to the entity.
description	A string. free-text description of the entity.
entityRelationship	A list of entity relationships (EntityRelationship [*]). A list of entities related to this entity.
entitySpecification	An entity specification reference (EntitySpecificationRef). reference to an EntitySpecification object.
isBundle	A boolean. isBundle determines whether an entity represents a single entity (false), or a bundle of entities(true).
isExternal	A boolean. isExternal determines whether an entity represents an external entity in the inventory.
lastUpdate	A date time (DateTime). Date and time of the last update of this REST resource.
lifecycleStatus	A string. Used to indicate the current lifecycle status of this managed entity.
name	A string. A string used to give a name to the entity.
note	A list of notes (Note [*]). Extra information about a given entity.
relatedParty	A list of related parties (RelatedParty [*]). Related Entity reference. A related party defines party or party role linked to a specific entity.
statusChangeDate	A date time (DateTime). A date time(DateTime). The date that the entity status

© TM Forum 2021. All Rights Reserved.

	changed to the current one.
validFor	A time period. The period for which this REST resource is valid.
version	A string. A field that identifies the specific version of an instance of an entity.

AttachmentRefOrValue sub-resource

An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
description	A string. A narrative text describing the content of the attachment.
href	An uri (Uri). URI for this Attachment.
id	A string. Unique identifier for this particular attachment.
url	An uri (Uri). Uniform Resource Locator, is a web page address (a subset of URI).
name	A string. The name of the attachment.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
attachmentType	A string. Attachment type such as video, picture.
content	A base 6 4 (Base64). The actual contents of the attachment object, if embedded, encoded as base64.
mimeType	A string. Attachment mime type such as extension file for video, picture and document.
size	A quantity (Quantity). The size of the attachment.
validFor	A time period. The period of time for which the attachment is valid.

Characteristic sub-resource

Describes a given characteristic of an object or entity through a name/value pair.

@baseType	A string. When sub-classing, this defines the super-class.

© TM Forum 2021. All Rights Reserved.

@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicRelationship	A list of characteristic relationships (CharacteristicRelationship [*]). Another Characteristic that is related to the current Characteristic;.
id	A string. Unique identifier of the characteristic.
name	A string. Name of the characteristic.
value	An any (Any). The value of the characteristic.
valueType	A string. Data type of the value of the characteristic.

CharacteristicRelationship sub-resource

Another Characteristic that is related to the current Characteristic;.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the characteristic.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. The type of relationship.

EntityRelationship sub-resource

A uni-directionmal relationship from this entity to a target entity instance.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
associationSpec	An association specification reference (AssociationSpecificationRef). reference to an AssociationSpecification object.
entity	An entity reference (EntityRef). The entity being referred to.

© TM Forum 2021. All Rights Reserved.

relationshipType	A string. Type of relationship such as containment, substitution, dependency, exclusivity.
role	A string. The association role for this entity.
validFor	A time period. The period for which the entityRelationship is valid.

Note sub-resource

Extra information about a given entity.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
author	A string. Author of the note.
date	A date time (DateTime). Date of the note.
id	A string. Identifier of the note within its containing entity.
text	A string. Text of the note.

Quantity sub-resource

An amount in a given unit.

amount	A float. Numeric value in a given unit.
units	A string. Unit.

RelatedParty sub-resource

Related Entity reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2021. All Rights Reserved.

@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

AssociationSpecificationRef relationship

reference to an AssociationSpecification object.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

EntityRef relationship

Entity reference schema to be use for all entityRef class.

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.

EntitySpecificationRef relationship

reference to an EntitySpecification object.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.

© TM Forum 2021. All Rights Reserved.

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. EntitySpecification version.

Json representation sample

Provided below is the json representation of an example of a 'ManagedEntity' resource object


```
{
  "id": "8951",
  "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8951",
  "name": "XYZ IOT Home Security Zone Hub",
  "description": "This entity represents an IOT Home Security Zone Hub instance",
  "version": "1.0",
  "validFor": {
    "startDateTime": "2021-03-20T00:00Z",
    "endDateTime": "2023-12-31T11:59:59Z"
  },
  "lastUpdate": "2021-03-20T00:00Z",
  "lifecycleStatus": "Active",
  "isBundle": false,
  "attachment": [
    {
      "description": "This attachment gives a user guide for setting of the home sucurity zone hub",
      "href": "https://mycsp.com:8080/tmf-api/documentManagement/v4/attachment/6899",
      "id": "6899",
      "url": "https://mycsp.com:7071/docloader?docnum=6899",
      "@attachmentType": "Document",
      "content": "",
      "mimeType": ".pdf",
      "validFor": {
        "startDateTime": "2021-03-20T00:00Z",
        "endDateTime": "2023-12-31T11:59:59Z"
      },
      "size": {
        "amount": 1400,
        "units": "KB"
      },
      "name": "XYZ Home Security Zone Hub Configuration",
      "@type": "Attachment",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Attachment.schema.json"
    }
```


© TM Forum 2021. All Rights Reserved.

```
  ],
  "characteristic": [
    {
      "name": "noOfZones",
      "id": "4421",
      "valueType": "number",
      "value": 4,
      "charSpecRelationship": [],
      "@type": "Characteristic",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
    },
    {
      "name": "ipAddress",
      "id": "4423",
      "valueType": "object",
      "value": {
        "address": "193.173.179.21",
        "netmask": "255.255.255.0",
        "@type": "IPAddressV4",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/IPAddressV4.schema.json"
      },
      "charSpecRelationship": [],
      "@type": "Characteristic",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
    },
    {
      "name": "serialNumber",
      "id": "4441",
      "valueType": "string",
      "value": "9753-1741-2321A",
      "charSpecRelationship": [],
      "@type": "Characteristic",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
    },
    {
      "name": "softwareVersion",
      "id": "4447",
      "valueType": "string",
      "value": "version 12.1",
      "charSpecRelationship": [],
      "@type": "Characteristic",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
    }
  ],
  "relatedParty": [
    {
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/individual/3023",
      "id": "3023",
      "name": "Gregory Peck",
```


© TM Forum 2021. All Rights Reserved.

```
        "role": "Customer",
        "@referredType": "PartyRole",
        "@type": "RelatedParty",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/EngagedParty/RelatedParty.schema.json"
      },
      {
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3023",
        "id": "3023",
        "name": "ABC Security Sensor Hub Supplier ",
        "role": "Supplier",
        "@referredType": "PartyRole",
        "@type": "RelatedParty",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/EngagedParty/RelatedParty.schema.json"
      },
      {
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3023",
        "id": "3023",
        "name": "XYZ Security Monitoring Co. ",
        "role": "Provider",
        "@referredType": "PartyRole",
        "@type": "RelatedParty",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/EngagedParty/RelatedParty.schema.json"
      }
    ],
    "entitySpecification": {
      "id": "6599",
      "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/entitySpecification/6599",
      "name": "IoT Security Device Specification",
      "@referredType": "ThingSpecification",
      "@type": "EntitySpecificationRef",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntitySpecificationRef.schema.json"
    },
    "entityRelationship": [
      {
        "relationshipType": "aggregates",
        "role": "hub",
        "entity": {
          "id": "8311",
          "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8887",
          "name": "Entrance Door Sensor",
          "@referredType": "IoTDevice",
          "@type": "EntityRef",
          "@baseType": "",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
        },
        "validFor": {
          "startDateTime": "2021-03-20T00:00Z",
          "endDateTime": "2023-12-31T11:59:59Z"
        },
```


© TM Forum 2021. All Rights Reserved.

```
      "associationSpec": {},
      "@type": "EntityRelationship",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
    },
    {
      "relationshipType": "aggregates",
      "role": "hub",
      "entity": {
        "id": "8317",
        "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8819",
        "name": "MotionSensor",
        "@referredType": "IoTDevice",
        "@type": "EntityRef",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
      },
      "validFor": {
        "startDateTime": "2021-03-20T00:00Z",
        "endDateTime": "2023-12-31T11:59:59Z"
      },
      "associationSpec": {},
      "@type": "EntityRelationship",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
    },
    {
      "relationshipType": "aggregates",
      "role": "hub",
      "entity": {
        "id": "8329",
        "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8821",
        "name": "Glass Break Sensor",
        "@referredType": "IoTDevice",
        "@type": "EntityRef",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
      },
      "validFor": {
        "startDateTime": "2021-03-20T00:00Z",
        "endDateTime": "2023-12-31T11:59:59Z"
      },
      "associationSpec": {},
      "@type": "EntityRelationship",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
    },
    {
      "relationshipType": "aggregates",
      "role": "hub",
      "entity": {
        "id": "8831",
        "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8831",
```


© TM Forum 2021. All Rights Reserved.

```
            "name": "HD Camera",
            "@referredType": "IoTDevice",
            "@type": "EntityRef",
            "@baseType": "",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
          },
          "validFor": {
            "startDateTime": "2021-03-20T00:00Z",
            "endDateTime": "2023-12-31T11:59:59Z"
          },
          "associationSpec": {},
          "@type": "EntityRelationship",
          "@baseType": "",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
        }
    ],
    "@type": "Thing",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/IoT/Thing.schema.json",
    "@baseType": "ManagedEntity"
}
```

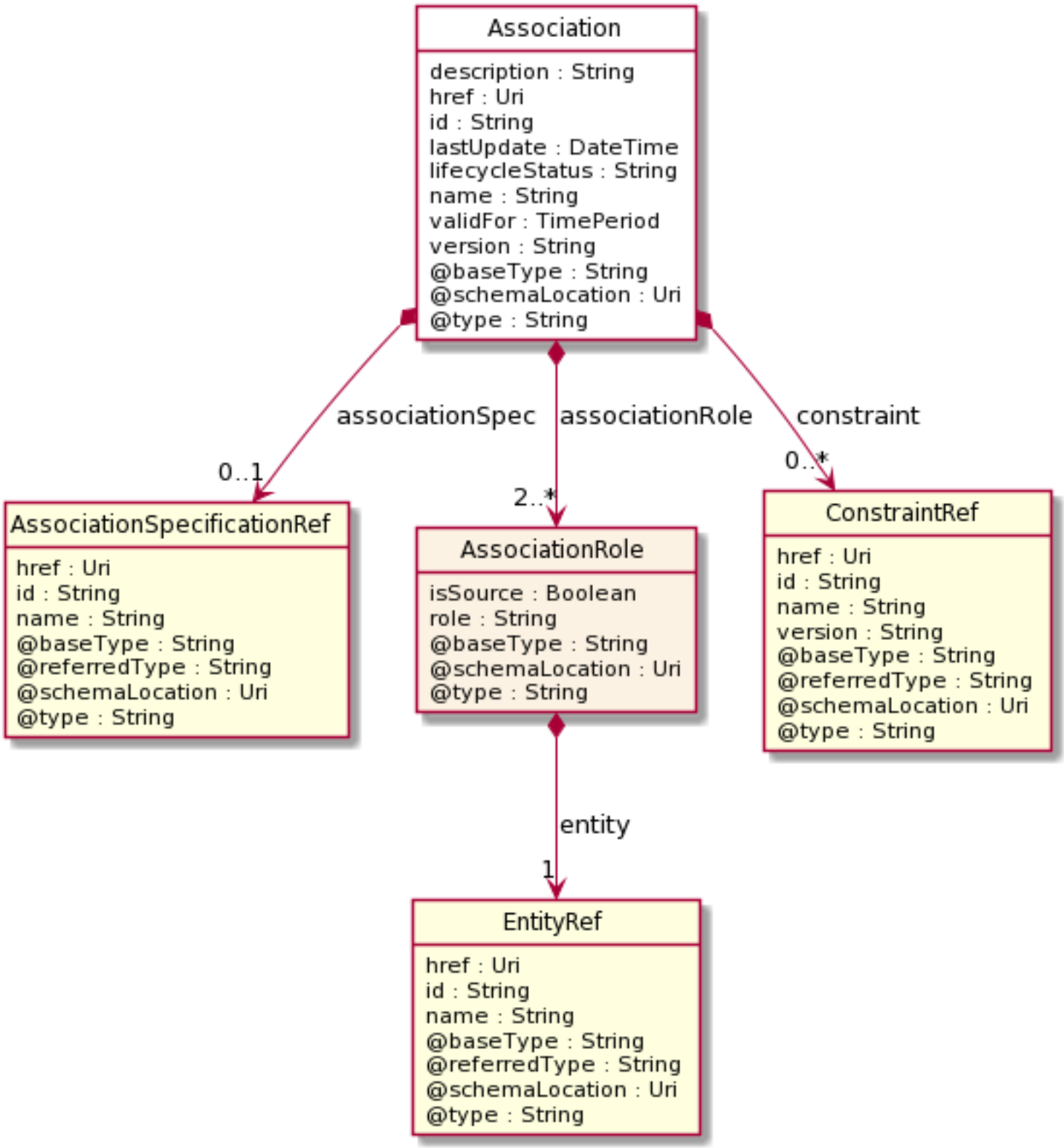

Association resource

Association is the class that describes a relationship between two or more entities or entity specifications based on a given association specification (relationship type). The role of each endpoint in the relationship is given by an association role. The type of endpoints in the relationship should match the ones as defined in the corresponding association role specification.

A relationship between entity/ entity specifications may be governed by conditions and rules which are addressed by constraint references in this resource. Constraints for a relationship may include new rules and conditions in addition to those defined for the corresponding association specification.

Resource model

© TM Forum 2021. All Rights Reserved.



© TM Forum 2021. All Rights Reserved.

Field descriptions

Association fields

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
associationRole	An association role (AssociationRole). The end point roles of this association.
associationSpec	An association specification reference (AssociationSpecificationRef). The specification of this association.
constraint	A list of constraint references (ConstraintRef [*]). Any constraints in this association.
description	A string. Description of the association.
lastUpdate	A date time (DateTime). The last modified date of this association object.
lifecycleStatus	A string. Indicates the current lifecycle status.
name	A string. Name given to this association.
validFor	A time period. The period for which this association is valid.
version	A string. Version of this REST resource.

AssociationRole sub-resource

This embedded object represents the role and type of each entity involved in a relationship.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

© TM Forum 2021. All Rights Reserved.

entity	An entity reference (EntityRef). The entity being referred to.
isSource	A boolean. A flag indicating if the participant involved in a relationship is the source or not. If this flag is true for both roles in an association, the association is bi-directional (both end points are navigable).
role	A string. The association role of this relationship participant as defined in the associationRoleSpecification.

AssociationSpecificationRef relationship

reference to an AssociationSpecification object.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

ConstraintRef relationship

Constraint reference. The Constraint resource represents a policy/rule applied to an entity or entity spec.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. constraint version.

© TM Forum 2021. All Rights Reserved.

EntityRef relationship

Entity reference schema to be use for all entityRef class.

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.

Json representation sample

Provided below is the json representation of an example of an 'Association' resource object


```
{
  "id": "17011",
  "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/association/17011",
  "name": "ReliesOn",
  "description": "An Uni-directional ReliesOn association from Electrical Fire Protection Managed Entity to Mechanical Fire Protection Managed Entity",
  "version": "1.0",
  "validFor": {
    "startDateTime": "2020-12-25T00:00Z",
    "endDateTime": "2023-12-31T11:59:59Z"
  },
  "lastUpdate": "2020-12-25T00:00Z",
  "lifecycleStatus": "Active",
  "associationSpec": {
    "id": "7727",
    "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/associationSpecification/7727",
    "name": "ReliesOn",
    "@referredType": "AssociationSpecfication",
    "@type": "AssociationSpecficationRef",
    "@baseType": "",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationSpecficationRef.schema.json"
  },
  "constraint": [
    {
      "href": "https://mycsp.com:8080/tmf-api/policyManagement/v4/policy/3947",
      "id": "3947",
      "name": "protectionDependencyRule",
      "version": "1.0",
      "@referredType": "Constraint",
```


© TM Forum 2021. All Rights Reserved.

```
        "@type": "ConstraintRef",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/ConstraintRef.schema.json"
      }
    ],
    "associationRole": [
      {
        "isSource": true,
        "role": "a_ReliesOn",
        "@type": "AssociationRole",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
        "@baseType": "",
        "entity": {
          "id": "18041",
          "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/managedEntity/18041",
          "name": "Fire Protection Electrical",
          "@type": "EntityRef",
          "@baseType": "",
          "referredType": "FireProtection",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
        }
      },
      {
        "isSource": false,
        "role": "z_ReliesOn",
        "@type": "AssociationRole",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
        "@baseType": "",
        "entity": {
          "id": "18013",
          "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/managedEntity/18013",
          "name": "Fire Protection Mechanical",
          "@type": "EntityRef",
          "@baseType": "",
          "referredType": "FireProtection",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
        }
      }
    ],
    "@type": "Association",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Association.schema.json",
    "@baseType": ""
}
```


Notification Resource Models

6 notifications are defined for this API

Notifications related to Entity:

- EntityCreateEvent

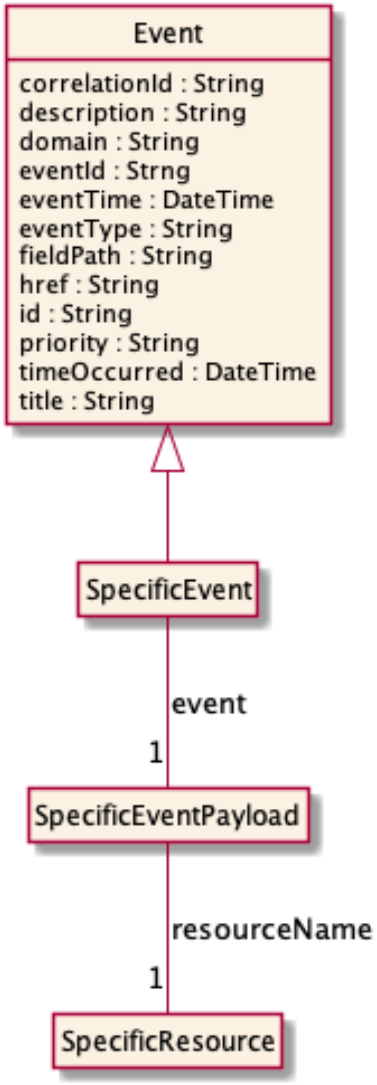
© TM Forum 2021. All Rights Reserved.

- EntityChangeEvent
- EntityDeleteEvent

Notifications related to Association:
- AssociationCreateEvent
- AssociationChangeEvent
- AssociationDeleteEvent

The notification structure for all notifications in this API follow the pattern depicted by the figure below.
A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType).

This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).



© TM Forum 2021. All Rights Reserved.

Entity Create Event

Notification EntityCreateEvent case for resource Entity

Json representation sample

Provided below is the json representation of an example of an 'EntityCreateEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"EntityCreateEvent",
    "event": {
      "entity" :
         {-- SEE Entity RESOURCE SAMPLE --}
   }
}
```

Entity Change Event

Notification EntityChangeEvent case for resource Entity

Json representation sample

Provided below is the json representation of an example of an 'EntityChangeEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"EntityChangeEvent",
    "event": {
      "entity" :
         {-- SEE Entity RESOURCE SAMPLE --}
   }
}
```

Entity Delete Event

Notification EntityDeleteEvent case for resource Entity

Json representation sample

Provided below is the json representation of an example of an 'EntityDeleteEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"EntityDeleteEvent",
    "event": {
      "entity" :
```

© TM Forum 2021. All Rights Reserved.

```
        {-- SEE Entity RESOURCE SAMPLE --}
    }
}
```

Association Create Event

Notification AssociationCreateEvent case for resource Association

Json representation sample

Provided below is the json representation of an example of an 'AssociationCreateEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"AssociationCreateEvent",
    "event": {
      "association" :
         {-- SEE Association RESOURCE SAMPLE --}
   }
}
```

Association Change Event

Notification AssociationChangeEvent case for resource Association

Json representation sample

Provided below is the json representation of an example of an 'AssociationChangeEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"AssociationChangeEvent",
    "event": {
      "association" :
         {-- SEE Association RESOURCE SAMPLE --}
   }
}
```

Association Delete Event

Notification AssociationDeleteEvent case for resource Association

© TM Forum 2021. All Rights Reserved.

Json representation sample

Provided below is the json representation of an example of an 'AssociationDeleteEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"AssociationDeleteEvent",
    "event": {
      "association" :
         {-- SEE Association RESOURCE SAMPLE --}
   }
}
```

© TM Forum 2021. All Rights Reserved.

API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines (TMF630).

Notifications are also described in a subsequent section.

Operations on Entity

List entities

```
GET /entity?fields=...&{filtering}
```

© TM Forum 2021. All Rights Reserved.

Description

This operation list entity entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving multiple active entities with the given type

Request

```
GET /tmf-api/entityInventory/v4/entity?@type=ToscaNode,Thing&lifecycleStatus=Active
Accept: application/json
```

Response

```
200

[
  {
    "id": "8887",
    "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8887",
    "name": "Virtual Network Function",
    "description": "This entity represents a Virtual Network Function instance",
    "version": "1.0",
    "validFor": {
      "startDateTime": "2021-03-20T00:00Z",
      "endDateTime": "2023-12-31T11:59:59Z"
    },
    "lastUpdate": "2021-03-20T00:00Z",
    "lifecycleStatus": "Active",
    "statusChangeDate": "2021-03-20T00:00Z",
    "isBundle": false,
    "isExternal": false,
    "context": "NFV",
    "note": [
      {
        "date": "2021-03-20T00:00Z",
        "id": "1",
        "author": "Victor Hugo",
        "text": "Some text"
      }
    ],
    "attachment": [
      {
        "description": "This attachment gives a description the software image which is directly loaded on the
```

© TM Forum 2021. All Rights Reserved.

```
virtualization container realizing this VNF",
            "href": "https://mycsp.com:8080/tmf-api/documentManagement/v4/attachment/6833",
            "id": "6833",
            "url": "https://mycsp.com:7070/docloader?docnum=6833",
            "@attachmentType": "Document",
            "content": "",
            "mimeType": ".pdf",
            "validFor": {
               "startDateTime": "2021-03-20T00:00Z",
               "endDateTime": "2023-12-31T11:59:59Z"
            },
            "size": {
               "amount": 5.7,
               "units": "MB"
            },
            "name": "VNF Features",
            "@type": "Attachment",
            "@baseType": "",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Attachment.schema.json"
         }
      ],
      "characteristic": [
         {
            "name": "Platform",
            "id": "4441",
            "valueType": "string",
            "value": "AWS",
            "charSpecRelationship": [],
            "@type": "Characteristic",
            "@baseType": "",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
         },
         {
            "name": "flavor_id",
            "id": "4451",
            "valueType": "string",
            "value": "Simple",
            "charSpecRelationship": [],
            "@type": "Characteristic",
            "@baseType": "",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
         },
         {
            "name": "autoScalable",
            "id": "4447",
            "valueType": "boolean",
            "value": true,
            "charSpecRelationship": [],
            "@type": "Characteristic",
            "@baseType": "",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
         }
      ],
```


© TM Forum 2021. All Rights Reserved.

```
    "relatedParty": [
      {
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3083",
        "id": "3083",
        "name": "Ericsson",
        "role": "Supplier",
        "@referredType": "PartyRole",
        "@type": "RelatedParty",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/EngagedParty/RelatedParty.schema.json"
      }
    ],
    "entitySpecification": {
      "id": "6599",
      "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/entitySpecification/6599",
      "name": "Tosca Node NFV Template",
      "@referredType": "ToscaNodeTemplate",
      "@type": "EntitySpecificationRef",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntitySpecificationRef.schema.json"
    },
    "entityRelationship": [
      {
        "relationshipType": "dependsOn",
        "role": "vnf",
        "entity": {
          "id": "8887",
          "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8887",
          "name": "VNFC A",
          "@referredType": "VnfcToscaNode",
          "@type": "EntityRef",
          "@baseType": "",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
        },
        "validFor": {
          "startDateTime": "2021-03-20T00:00Z",
          "endDateTime": "2023-12-31T11:59:59Z"
        },
        "associationSpec": {},
        "@type": "EntityRelationship",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
      },
      {
        "relationshipType": "requires",
        "role": "vnf",
        "entity": {
          "id": "8819",
          "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8819",
          "name": "Virtual Deployment A",
          "@referredType": "VDToscaNode",
          "@type": "EntityRef",
          "@baseType": "",
```


© TM Forum 2021. All Rights Reserved.

```
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
          },
          "validFor": {
            "startDateTime": "2021-03-20T00:00Z",
            "endDateTime": "2023-12-31T11:59:59Z"
          },
          "associationSpec": {},
          "@type": "EntityRelationship",
          "@baseType": "",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
        },
        {
          "relationshipType": "requires",
          "role": "vnf",
          "entity": {
            "id": "8821",
            "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8821",
            "name": "Virtual Link A",
            "@referredType": "NsVirtualLinkToscaNode",
            "@type": "EntityRef",
            "@baseType": "",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
          },
          "validFor": {
            "startDateTime": "2021-03-20T00:00Z",
            "endDateTime": "2023-12-31T11:59:59Z"
          },
          "associationSpec": {},
          "@type": "EntityRelationship",
          "@baseType": "",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
        },
        {
          "relationshipType": "requires",
          "role": "vnf",
          "entity": {
            "id": "8831",
            "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8831",
            "name": "Connection Point CP1",
            "@referredType": "CPToscaNode",
            "@type": "EntityRef",
            "@baseType": "",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
          },
          "validFor": {
            "startDateTime": "2021-03-20T00:00Z",
            "endDateTime": "2023-12-31T11:59:59Z"
          },
          "associationSpec": {},
          "@type": "EntityRelationship",
          "@baseType": "",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
        }
```


© TM Forum 2021. All Rights Reserved.

```
    ],
    "@type": "ToscaNode",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/ToscaNode.schema.json",
    "@baseType": "ManagedEntity"
  },
  {
    "id": "8951",
    "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8951",
    "name": "XYZ IOT Home Security Zone Hub",
    "description": "This entity represents an IOT Home Security Zone Hub instance",
    "version": "1.0",
    "validFor": {
      "startDateTime": "2021-03-20T00:00Z",
      "endDateTime": "2023-12-31T11:59:59Z"
    },
    "lastUpdate": "2021-03-20T00:00Z",
    "lifecycleStatus": "Active",
    "statusChangeDate": "2021-03-20T00:00Z",
    "isBundle": false,
    "isExternal": false,
    "context": "IOT",
    "note": [
      {
        "date": "2021-03-20T00:00Z",
        "id": "1",
        "author": "Victor Hugo",
        "text": "Some text"
      }
    ],
    "attachment": [
      {
        "description": "This attachment gives a user guide for setting of the home sucurity zone hub",
        "href": "https://mycsp.com:8080/tmf-api/documentManagement/v4/attachment/6899",
        "id": "6899",
        "url": "https://mycsp.com:7071/docloader?docnum=6899",
        "@attachmentType": "Document",
        "content": "",
        "mimeType": ".pdf",
        "validFor": {
          "startDateTime": "2021-03-20T00:00Z",
          "endDateTime": "2023-12-31T11:59:59Z"
        },
        "size": {
          "amount": 1400,
          "units": "KB"
        },
        "name": "XYZ Home Security Zone Hub Configuration",
        "@type": "Attachment",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Attachment.schema.json"
      }
    ],
    "characteristic": [
```


© TM Forum 2021. All Rights Reserved.

```
        {
          "name": "noOfZones",
          "id": "4421",
          "valueType": "number",
          "value": 4,
          "charSpecRelationship": [],
          "@type": "Characteristic",
          "@baseType": "",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
        },
        {
          "name": "ipAddress",
          "id": "4423",
          "valueType": "object",
          "value": {
            "address": "193.173.179.21",
            "netmask": "255.255.255.0",
            "@type": "IPAddressV4",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/IPAddressV4.schema.json"
          },
          "charSpecRelationship": [],
          "@type": "Characteristic",
          "@baseType": "",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
        },
        {
          "name": "serialNumber",
          "id": "4441",
          "valueType": "string",
          "value": "9753-1741-2321A",
          "charSpecRelationship": [],
          "@type": "Characteristic",
          "@baseType": "",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
        },
        {
          "name": "softwareVersion",
          "id": "4447",
          "valueType": "string",
          "value": "version 12.1",
          "charSpecRelationship": [],
          "@type": "Characteristic",
          "@baseType": "",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
        }
      ],
      "relatedParty": [
        {
          "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/individual/3023",
          "id": "3023",
          "name": "Gregory Peck",
          "role": "Customer",
          "@referredType": "PartyRole",
```


© TM Forum 2021. All Rights Reserved.

```
        "@type": "RelatedParty",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/EngagedParty/RelatedParty.schema.json"
      },
      {
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3023",
        "id": "3023",
        "name": "ABC Security Sensor Hub Supplier ",
        "role": "Supplier",
        "@referredType": "PartyRole",
        "@type": "RelatedParty",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/EngagedParty/RelatedParty.schema.json"
      },
      {
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3023",
        "id": "3023",
        "name": "XYZ Security Monitoring Co. ",
        "role": "Provider",
        "@referredType": "PartyRole",
        "@type": "RelatedParty",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/EngagedParty/RelatedParty.schema.json"
      }
    ],
    "entitySpecification": {
      "id": "6599",
      "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/entitySpecification/6599",
      "name": "IoT Security Device Specification",
      "@referredType": "ThingSpecification",
      "@type": "EntitySpecificationRef",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntitySpecificationRef.schema.json"
    },
    "entityRelationship": [
      {
        "relationshipType": "aggregates",
        "role": "hub",
        "entity": {
          "id": "8311",
          "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8887",
          "name": "Entrance Door Sensor",
          "@referredType": "IoTDevice",
          "@type": "EntityRef",
          "@baseType": "",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
        },
        "validFor": {
          "startDateTime": "2021-03-20T00:00Z",
          "endDateTime": "2023-12-31T11:59:59Z"
        },
        "associationSpec": {},
        "@type": "EntityRelationship",
```


© TM Forum 2021. All Rights Reserved.

```
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
    },
    {
      "relationshipType": "aggregates",
      "role": "hub",
      "entity": {
        "id": "8317",
        "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8819",
        "name": "MotionSensor",
        "@referredType": "IoTDevice",
        "@type": "EntityRef",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
      },
      "validFor": {
        "startDateTime": "2021-03-20T00:00Z",
        "endDateTime": "2023-12-31T11:59:59Z"
      },
      "associationSpec": {},
      "@type": "EntityRelationship",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
    },
    {
      "relationshipType": "aggregates",
      "role": "hub",
      "entity": {
        "id": "8329",
        "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8821",
        "name": "Glass Break Sensor",
        "@referredType": "IoTDevice",
        "@type": "EntityRef",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
      },
      "validFor": {
        "startDateTime": "2021-03-20T00:00Z",
        "endDateTime": "2023-12-31T11:59:59Z"
      },
      "associationSpec": {},
      "@type": "EntityRelationship",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
    },
    {
      "relationshipType": "aggregates",
      "role": "hub",
      "entity": {
        "id": "8831",
        "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8831",
        "name": "HD Camera",
        "@referredType": "IoTDevice",
```


© TM Forum 2021. All Rights Reserved.

```
            "@type": "EntityRef",
            "@baseType": "",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
          },
          "validFor": {
            "startDateTime": "2021-03-20T00:00Z",
            "endDateTime": "2023-12-31T11:59:59Z"
          },
          "associationSpec": {},
          "@type": "EntityRelationship",
          "@baseType": "",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
        }
    ],
    "@type": "Thing",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/IoT/Thing.schema.json",
    "@baseType": "ManagedEntity"
  }
]
```


Retrieve entity

`GET /entity/{id}?fields=...&{filtering}`

Description

This operation retrieves an entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving a specific entity

Request
GET /tmf-api/entityInventory/v4/entity/8951 Accept: application/json
Response
200 { "id": "8951", "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8951",

© TM Forum 2021. All Rights Reserved.

```
"name": "XYZ IOT Home Security Zone Hub",
"description": "This entity represents an IOT Home Security Zone Hub instance",
"version": "1.0",
"validFor": {
   "startDateTime": "2021-03-20T00:00Z",
   "endDateTime": "2023-12-31T11:59:59Z"
},
"lastUpdate": "2021-03-20T00:00Z",
"lifecycleStatus": "Active",
"statusChangeDate": "2021-03-20T00:00Z",
"isBundle": false,
"isExternal": false,
"context": "IOT",
"note": [
  {
     "date": "2021-03-20T00:00Z",
     "id": "1",
     "author": "Victor Hugo",
     "text": "Some text"
  }
],
"attachment": [
  {
     "description": "This attachment gives a user guide for setting of the home sucurity zone hub",
     "href": "https://mycsp.com:8080/tmf-api/documentManagement/v4/attachment/6899",
     "id": "6899",
     "url": "https://mycsp.com:7071/docloader?docnum=6899",
     "@attachmentType": "Document",
     "content": "",
     "mimeType": ".pdf",
     "validFor": {
        "startDateTime": "2021-03-20T00:00Z",
        "endDateTime": "2023-12-31T11:59:59Z"
     },
     "size": {
        "amount": 1400,
        "units": "KB"
     },
     "name": "XYZ Home Security Zone Hub Configuration",
     "@type": "Attachment",
     "@baseType": "",
     "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Attachment.schema.json"
  }
],
"characteristic": [
  {
     "name": "noOfZones",
     "id": "4421",
     "valueType": "number",
     "value": 4,
     "charSpecRelationship": [],
     "@type": "Characteristic",
     "@baseType": "",
```


© TM Forum 2021. All Rights Reserved.

```
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
      },
      {
        "name": "ipAddress",
        "id": "4423",
        "valueType": "object",
        "value": {
          "address": "193.173.179.21",
          "netmask": "255.255.255.0",
          "@type": "IPAddressV4",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/IPAddressV4.schema.json"
        },
        "charSpecRelationship": [],
        "@type": "Characteristic",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
      },
      {
        "name": "serialNumber",
        "id": "4441",
        "valueType": "string",
        "value": "9753-1741-2321A",
        "charSpecRelationship": [],
        "@type": "Characteristic",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
      },
      {
        "name": "softwareVersion",
        "id": "4447",
        "valueType": "string",
        "value": "version 12.1",
        "charSpecRelationship": [],
        "@type": "Characteristic",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
      }
    ],
    "relatedParty": [
      {
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/individual/3023",
        "id": "3023",
        "name": "Gregory Peck",
        "role": "Customer",
        "@referredType": "PartyRole",
        "@type": "RelatedParty",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/EngagedParty/RelatedParty.schema.json"
      },
      {
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3023",
        "id": "3023",
        "name": "ABC Security Sensor Hub Supplier ",
```


© TM Forum 2021. All Rights Reserved.

```
        "role": "Supplier",
        "@referredType": "PartyRole",
        "@type": "RelatedParty",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/EngagedParty/RelatedParty.schema.json"
      },
      {
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3023",
        "id": "3023",
        "name": "XYZ Security Monitoring Co. ",
        "role": "Provider",
        "@referredType": "PartyRole",
        "@type": "RelatedParty",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/EngagedParty/RelatedParty.schema.json"
      }
    ],
    "entitySpecification": {
      "id": "6599",
      "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/entitySpecification/6599",
      "name": "IoT Security Device Specification",
      "@referredType": "ThingSpecification",
      "@type": "EntitySpecificationRef",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntitySpecificationRef.schema.json"
    },
    "entityRelationship": [
      {
        "relationshipType": "aggregates",
        "role": "hub",
        "entity": {
          "id": "8311",
          "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8887",
          "name": "Entrance Door Sensor",
          "@referredType": "IoTDevice",
          "@type": "EntityRef",
          "@baseType": "",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
        },
        "validFor": {
          "startDateTime": "2021-03-20T00:00Z",
          "endDateTime": "2023-12-31T11:59:59Z"
        },
        "associationSpec": {},
        "@type": "EntityRelationship",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
      },
      {
        "relationshipType": "aggregates",
        "role": "hub",
        "entity": {
          "id": "8317",
```


© TM Forum 2021. All Rights Reserved.

```
        "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8819",
        "name": "MotionSensor",
        "@referredType": "IoTDevice",
        "@type": "EntityRef",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
      },
      "validFor": {
        "startDateTime": "2021-03-20T00:00Z",
        "endDateTime": "2023-12-31T11:59:59Z"
      },
      "associationSpec": {},
      "@type": "EntityRelationship",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
    },
    {
      "relationshipType": "aggregates",
      "role": "hub",
      "entity": {
        "id": "8329",
        "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8821",
        "name": "Glass Break Sensor",
        "@referredType": "IoTDevice",
        "@type": "EntityRef",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
      },
      "validFor": {
        "startDateTime": "2021-03-20T00:00Z",
        "endDateTime": "2023-12-31T11:59:59Z"
      },
      "associationSpec": {},
      "@type": "EntityRelationship",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
    },
    {
      "relationshipType": "aggregates",
      "role": "hub",
      "entity": {
        "id": "8831",
        "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8831",
        "name": "HD Camera",
        "@referredType": "IoTDevice",
        "@type": "EntityRef",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
      },
      "validFor": {
        "startDateTime": "2021-03-20T00:00Z",
        "endDateTime": "2023-12-31T11:59:59Z"
      },
```


© TM Forum 2021. All Rights Reserved.

```
        "associationSpec": {},
        "@type": "EntityRelationship",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
      }
  ],
  "@type": "Thing",
  "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/IoT/Thing.schema.json",
  "@baseType": "ManagedEntity"
}
```

Create entity

POST /entity

Description

This operation creates an entity entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a Entity, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

Mandatory Attributes	Rule
@type	

Non Mandatory Attributes	Rule
@baseType	
@schemaLocation	

Usage Samples

Here's an example of a request for creating an managed entity.

Request

```
POST /tmf-api/entityInventory/v4/entity
Content-Type: application/json

{
  "name": "XYZ IOT Home Security Zone Hub",
  "description": "This entity represents an IOT Home Security Zone Hub instance",
  "version": "1.0",
  "validFor": {
```

© TM Forum 2021. All Rights Reserved.

```
    "startDateTime": "2021-03-20T00:00Z",
    "endDateTime": "2023-12-31T11:59:59Z"
  },
  "lifecycleStatus": "Active",
  "isBundle": false,
  "isExternal": false,
  "context": "IOT",
  "note": [
    {
      "date": "2021-03-20T00:00Z",
      "id": "1",
      "author": "Victor Hugo",
      "text": "Some text"
    }
  ],
  "attachment": [
    {
      "description": "This attachment gives a user guide for setting of the home sucurity zone hub",
      "href": "https://mycsp.com:8080/tmf-api/documentManagement/v4/attachment/6899",
      "id": "6899",
      "url": "https://mycsp.com:7071/docloader?docnum=6899",
      "@attachmentType": "Document",
      "mimeType": ".pdf",
      "validFor": {
        "startDateTime": "2021-03-20T00:00Z",
        "endDateTime": "2023-12-31T11:59:59Z"
      },
      "size": {
        "amount": 1400,
        "units": "KB"
      },
      "name": "XYZ Home Security Zone Hub Configuration",
      "@type": "Attachment",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Attachment.schema.json"
    }
  ],
  "characteristic": [
    {
      "name": "noOfZones",
      "id": "4421",
      "valueType": "number",
      "value": 4,
      "charSpecRelationship": [],
      "@type": "Characteristic",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
    },
    {
      "name": "ipAddress",
      "id": "4423",
      "valueType": "object",
      "value": {
        "address": "193.173.179.21",
        "netmask": "255.255.255.0",
```


© TM Forum 2021. All Rights Reserved.

```
            "@type": "IPAddressV4",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/IPAddressV4.schema.json"
          },
          "charSpecRelationship": [],
          "@type": "Characteristic",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
        },
        {
          "name": "serialNumber",
          "id": "4441",
          "valueType": "string",
          "value": "9753-1741-2321A",
          "charSpecRelationship": [],
          "@type": "Characteristic",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
        },
        {
          "name": "softwareVersion",
          "id": "4447",
          "valueType": "string",
          "value": "version 12.1",
          "charSpecRelationship": [],
          "@type": "Characteristic",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
        }
      ],
      "relatedParty": [
        {
          "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/individual/3023",
          "id": "3023",
          "name": "Gregory Peck",
          "role": "Customer",
          "@referredType": "PartyRole",
          "@type": "RelatedParty",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/EngagedParty/RelatedParty.schema.json"
        },
        {
          "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3023",
          "id": "3023",
          "name": "ABC Security Sensor Hub Supplier ",
          "role": "Supplier",
          "@referredType": "PartyRole",
          "@type": "RelatedParty",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/EngagedParty/RelatedParty.schema.json"
        },
        {
          "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3023",
          "id": "3023",
          "name": "XYZ Security Monitoring Co. ",
          "role": "Provider",
          "@referredType": "PartyRole",
          "@type": "RelatedParty",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/EngagedParty/RelatedParty.schema.json"
```


© TM Forum 2021. All Rights Reserved.

```
    }
  ],
  "entitySpecification": {
    "id": "6599",
    "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/entitySpecification/6599",
    "name": "IoT Security Device Specification",
    "@referredType": "ThingSpecification",
    "@type": "EntitySpecificationRef",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntitySpecificationRef.schema.json"
  },
  "entityRelationship": [
    {
      "relationshipType": "aggregates",
      "role": "hub",
      "entity": {
        "id": "8311",
        "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8887",
        "name": "Entrance Door Sensor",
        "@referredType": "IoTDevice",
        "@type": "EntityRef",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
      },
      "validFor": {
        "startDateTime": "2021-03-20T00:00Z",
        "endDateTime": "2023-12-31T11:59:59Z"
      },
      "associationSpec": {},
      "@type": "EntityRelationship",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
    },
    {
      "relationshipType": "aggregates",
      "role": "hub",
      "entity": {
        "id": "8317",
        "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8819",
        "name": "MotionSensor",
        "@referredType": "IoTDevice",
        "@type": "EntityRef",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
      },
      "validFor": {
        "startDateTime": "2021-03-20T00:00Z",
        "endDateTime": "2023-12-31T11:59:59Z"
      },
      "associationSpec": {},
      "@type": "EntityRelationship",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
    },
    {
      "relationshipType": "aggregates",
      "role": "hub",
      "entity": {
```


© TM Forum 2021. All Rights Reserved.

```
            "id": "8329",
            "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8821",
            "name": "Glass Break Sensor",
            "@referredType": "IoTDevice",
            "@type": "EntityRef",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
          },
          "validFor": {
            "startDateTime": "2021-03-20T00:00Z",
            "endDateTime": "2023-12-31T11:59:59Z"
          },
          "associationSpec": {},
          "@type": "EntityRelationship",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
        },
        {
          "relationshipType": "aggregates",
          "role": "hub",
          "entity": {
            "id": "8831",
            "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8831",
            "name": "HD Camera",
            "@referredType": "IoTDevice",
            "@type": "EntityRef",
            "@baseType": "",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
          },
          "validFor": {
            "startDateTime": "2021-03-20T00:00Z",
            "endDateTime": "2023-12-31T11:59:59Z"
          },
          "associationSpec": {},
          "@type": "EntityRelationship",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
        }
      ],
      "@type": "IoTDevice",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/IoT/IoTDevice.schema.json",
      "@baseType": "ManagedEntity"
}
```


Response

201


```
{
    "id": "8951",
    "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8951",
    "name": "XYZ IOT Home Security Zone Hub",
    "description": "This entity represents an IOT Home Security Zone Hub instance",
    "version": "1.0",
```


© TM Forum 2021. All Rights Reserved.

```
  "validFor": {
     "startDateTime": "2021-03-20T00:00Z",
     "endDateTime": "2023-12-31T11:59:59Z"
  },
  "lastUpdate": "2021-03-20T00:00Z",
  "lifecycleStatus": "Active",
  "statusChangeDate": "2021-03-20T00:00Z",
  "isBundle": false,
  "isExternal": false,
  "context": "IOT",
  "note": [
     {
        "date": "2021-03-20T00:00Z",
        "id": "1",
        "author": "Victor Hugo",
        "text": "Some text"
     }
  ],
  "attachment": [
     {
        "description": "This attachment gives a user guide for setting of the home sucurity zone hub",
        "href": "https://mycsp.com:8080/tmf-api/documentManagement/v4/attachment/6899",
        "id": "6899",
        "url": "https://mycsp.com:7071/docloader?docnum=6899",
        "@attachmentType": "Document",
        "content": "",
        "mimeType": ".pdf",
        "validFor": {
           "startDateTime": "2021-03-20T00:00Z",
           "endDateTime": "2023-12-31T11:59:59Z"
        },
        "size": {
           "amount": 1400,
           "units": "KB"
        },
        "name": "XYZ Home Security Zone Hub Configuration",
        "@type": "Attachment",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Attachment.schema.json"
     }
  ],
  "characteristic": [
     {
        "name": "noOfZones",
        "id": "4421",
        "valueType": "number",
        "value": 4,
        "charSpecRelationship": [],
        "@type": "Characteristic",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
     },
     {
```


© TM Forum 2021. All Rights Reserved.

```
        "name": "ipAddress",
        "id": "4423",
        "valueType": "object",
        "value": {
          "address": "193.173.179.21",
          "netmask": "255.255.255.0",
          "@type": "IPAddressV4",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/IPAddressV4.schema.json"
        },
        "charSpecRelationship": [],
        "@type": "Characteristic",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
      },
      {
        "name": "serialNumber",
        "id": "4441",
        "valueType": "string",
        "value": "9753-1741-2321A",
        "charSpecRelationship": [],
        "@type": "Characteristic",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
      },
      {
        "name": "softwareVersion",
        "id": "4447",
        "valueType": "string",
        "value": "version 12.1",
        "charSpecRelationship": [],
        "@type": "Characteristic",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
      }
    ],
    "relatedParty": [
      {
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/individual/3023",
        "id": "3023",
        "name": "Gregory Peck",
        "role": "Customer",
        "@referredType": "PartyRole",
        "@type": "RelatedParty",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/EngagedParty/RelatedParty.schema.json"
      },
      {
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3023",
        "id": "3023",
        "name": "ABC Security Sensor Hub Supplier ",
        "role": "Supplier",
        "@referredType": "PartyRole",
        "@type": "RelatedParty",
```


© TM Forum 2021. All Rights Reserved.

```
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/EngagedParty/RelatedParty.schema.json"
      },
      {
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3023",
        "id": "3023",
        "name": "XYZ Security Monitoring Co. ",
        "role": "Provider",
        "@referredType": "PartyRole",
        "@type": "RelatedParty",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/EngagedParty/RelatedParty.schema.json"
      }
    ],
    "entitySpecification": {
      "id": "6599",
      "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/entitySpecification/6599",
      "name": "IoT Security Device Specification",
      "@referredType": "ThingSpecification",
      "@type": "EntitySpecificationRef",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntitySpecificationRef.schema.json"
    },
    "entityRelationship": [
      {
        "relationshipType": "aggregates",
        "role": "hub",
        "entity": {
          "id": "8311",
          "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8887",
          "name": "Entrance Door Sensor",
          "@referredType": "IoTDevice",
          "@type": "EntityRef",
          "@baseType": "",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
        },
        "validFor": {
          "startDateTime": "2021-03-20T00:00Z",
          "endDateTime": "2023-12-31T11:59:59Z"
        },
        "associationSpec": {},
        "@type": "EntityRelationship",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
      },
      {
        "relationshipType": "aggregates",
        "role": "hub",
        "entity": {
          "id": "8317",
          "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8819",
          "name": "MotionSensor",
          "@referredType": "IoTDevice",
```


© TM Forum 2021. All Rights Reserved.

```
        "@type": "EntityRef",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
      },
      "validFor": {
        "startDateTime": "2021-03-20T00:00Z",
        "endDateTime": "2023-12-31T11:59:59Z"
      },
      "associationSpec": {},
      "@type": "EntityRelationship",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
    },
    {
      "relationshipType": "aggregates",
      "role": "hub",
      "entity": {
        "id": "8329",
        "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8821",
        "name": "Glass Break Sensor",
        "@referredType": "IoTDevice",
        "@type": "EntityRef",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
      },
      "validFor": {
        "startDateTime": "2021-03-20T00:00Z",
        "endDateTime": "2023-12-31T11:59:59Z"
      },
      "associationSpec": {},
      "@type": "EntityRelationship",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
    },
    {
      "relationshipType": "aggregates",
      "role": "hub",
      "entity": {
        "id": "8831",
        "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8831",
        "name": "HD Camera",
        "@referredType": "IoTDevice",
        "@type": "EntityRef",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
      },
      "validFor": {
        "startDateTime": "2021-03-20T00:00Z",
        "endDateTime": "2023-12-31T11:59:59Z"
      },
      "associationSpec": {},
      "@type": "EntityRelationship",
      "@baseType": "",
```


© TM Forum 2021. All Rights Reserved.

```
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
      }
    ],
    "@type": "Thing",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/IoT/Thing.schema.json",
    "@baseType": "ManagedEntity"
}
```


Patch entity

PATCH /entity/{id}

Description

This operation allows partial updates of an entity entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional.

Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

The tables below provide the list of patchable and non patchable attributes, including constraint rules on their usage.

Patchable Attributes	Rule

Non Patchable Attributes	Rule
id	
href	
@type	
@baseType	
@schemaLocation	

Usage Samples

Here's an example of a request for patching an managed entity. In this example, a new version is set, together with a new validity period

Request
PATCH /tmf-api/entityInventory/v4/entity/8951 Content-Type: application/merge-patch+json

© TM Forum 2021. All Rights Reserved.

```
{
   "version": "2.0",
   "validFor": {
      "startDateTime": "2021-03-20T00:00Z",
      "endDateTime": "2022-12-31T11:59:59Z"
   }
}
```


Response

200


```
{
   "id": "8951",
   "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8951",
   "name": "XYZ IOT Home Security Zone Hub",
   "description": "This entity represents an IOT Home Security Zone Hub instance",
   "version": "2.0",
   "validFor": {
      "startDateTime": "2021-03-20T00:00Z",
      "endDateTime": "2022-12-31T11:59:59Z"
   },
   "lastUpdate": "2021-05-17T00:00Z",
   "lifecycleStatus": "Active",
   "statusChangeDate": "2021-03-20T00:00Z",
   "isBundle": false,
   "isExternal": false,
   "context": "IOT",
   "note": [
      {
         "date": "2021-03-20T00:00Z",
         "id": "1",
         "author": "Victor Hugo",
         "text": "Some text"
      }
   ],
   "attachment": [
      {
         "description": "This attachment gives a user guide for setting of the home sucurity zone hub",
         "href": "https://mycsp.com:8080/tmf-api/documentManagement/v4/attachment/6899",
         "id": "6899",
         "url": "https://mycsp.com:7071/docloader?docnum=6899",
         "@attachmentType": "Document",
         "content": "",
         "mimeType": ".pdf",
         "validFor": {
            "startDateTime": "2021-03-20T00:00Z",
            "endDateTime": "2023-12-31T11:59:59Z"
         },
         "size": {
```


© TM Forum 2021. All Rights Reserved.

```
            "amount": 1400,
            "units": "KB"
          },
          "name": "XYZ Home Security Zone Hub Configuration",
          "@type": "Attachment",
          "@baseType": "",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Attachment.schema.json"
       }
    ],
    "characteristic": [
       {
          "name": "noOfZones",
          "id": "4421",
          "valueType": "number",
          "value": 4,
          "charSpecRelationship": [],
          "@type": "Characteristic",
          "@baseType": "",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
       },
       {
          "name": "ipAddress",
          "id": "4423",
          "valueType": "object",
          "value": {
             "address": "193.173.179.21",
             "netmask": "255.255.255.0",
             "@type": "IPAddressV4",
             "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/IPAddressV4.schema.json"
          },
          "charSpecRelationship": [],
          "@type": "Characteristic",
          "@baseType": "",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
       },
       {
          "name": "serialNumber",
          "id": "4441",
          "valueType": "string",
          "value": "9753-1741-2321A",
          "charSpecRelationship": [],
          "@type": "Characteristic",
          "@baseType": "",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
       },
       {
          "name": "softwareVersion",
          "id": "4447",
          "valueType": "string",
          "value": "version 12.1",
          "charSpecRelationship": [],
          "@type": "Characteristic",
          "@baseType": "",
```


© TM Forum 2021. All Rights Reserved.

```
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Characteristic.schema.json"
      }
    ],
    "relatedParty": [
      {
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/individual/3023",
        "id": "3023",
        "name": "Gregory Peck",
        "role": "Customer",
        "@referredType": "PartyRole",
        "@type": "RelatedParty",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/EngagedParty/RelatedParty.schema.json"
      },
      {
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3023",
        "id": "3023",
        "name": "ABC Security Sensor Hub Supplier ",
        "role": "Supplier",
        "@referredType": "PartyRole",
        "@type": "RelatedParty",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/EngagedParty/RelatedParty.schema.json"
      },
      {
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3023",
        "id": "3023",
        "name": "XYZ Security Monitoring Co. ",
        "role": "Provider",
        "@referredType": "PartyRole",
        "@type": "RelatedParty",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/EngagedParty/RelatedParty.schema.json"
      }
    ],
    "entitySpecification": {
      "id": "6599",
      "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/entitySpecification/6599",
      "name": "IoT Security Device Specification",
      "@referredType": "ThingSpecification",
      "@type": "EntitySpecificationRef",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntitySpecificationRef.schema.json"
    },
    "entityRelationship": [
      {
        "relationshipType": "aggregates",
        "role": "hub",
        "entity": {
          "id": "8311",
          "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8887",
          "name": "Entrance Door Sensor",
          "@referredType": "IoTDevice",
```


© TM Forum 2021. All Rights Reserved.

```
            "@type": "EntityRef",
            "@baseType": "",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
          },
          "validFor": {
            "startDateTime": "2021-03-20T00:00Z",
            "endDateTime": "2023-12-31T11:59:59Z"
          },
          "associationSpec": {},
          "@type": "EntityRelationship",
          "@baseType": "",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
        },
        {
          "relationshipType": "aggregates",
          "role": "hub",
          "entity": {
            "id": "8317",
            "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8819",
            "name": "MotionSensor",
            "@referredType": "IoTDevice",
            "@type": "EntityRef",
            "@baseType": "",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
          },
          "validFor": {
            "startDateTime": "2021-03-20T00:00Z",
            "endDateTime": "2023-12-31T11:59:59Z"
          },
          "associationSpec": {},
          "@type": "EntityRelationship",
          "@baseType": "",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
        },
        {
          "relationshipType": "aggregates",
          "role": "hub",
          "entity": {
            "id": "8329",
            "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8821",
            "name": "Glass Break Sensor",
            "@referredType": "IoTDevice",
            "@type": "EntityRef",
            "@baseType": "",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
          },
          "validFor": {
            "startDateTime": "2021-03-20T00:00Z",
            "endDateTime": "2023-12-31T11:59:59Z"
          },
          "associationSpec": {},
          "@type": "EntityRelationship",
          "@baseType": "",
```


© TM Forum 2021. All Rights Reserved.

```
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
      },
      {
        "relationshipType": "aggregates",
        "role": "hub",
        "entity": {
          "id": "8831",
          "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/entity/8831",
          "name": "HD Camera",
          "@referredType": "IoTDevice",
          "@type": "EntityRef",
          "@baseType": "",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
        },
        "validFor": {
          "startDateTime": "2021-03-20T00:00Z",
          "endDateTime": "2023-12-31T11:59:59Z"
        },
        "associationSpec": {},
        "@type": "EntityRelationship",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRelationship.schema.json"
      }
    ],
    "@type": "Thing",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/IoT/Thing.schema.json",
    "@baseType": "ManagedEntity"
}
```


Delete entity

`DELETE /entity/{id}`

Description

This operation deletes an entity entity.

Usage Samples

Here's an example of a request for deleting a given entity.

Request
DELETE /tmf-api/entityInventory/v4/entity/8951
Response

© TM Forum 2021. All Rights Reserved.

204

Operations on Managed Entity

Operations on Association

List associations

GET /association?fields=...&{filtering}

Description

This operation list association entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving multiple associations.

Request
GET /tmf-api/entityInventory/v4/association?lifecycleStatus=Active Accept: application/json
Response

200
 	
[
{
"id": "17011",
"href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/association/17011",
"name": "ReliesOn",
"description": "An Uni-directional ReliesOn association from Electrical Fire Protection Managed Entity to Mechanical Fire Protection Managed Entity",
"version": "1.0",
"validFor": {
"startDateTime": "2020-12-25T00:00Z",
"endDateTime": "2023-12-31T11:59:59Z"
},
"lastUpdate": "2020-12-25T00:00Z",
"lifecycleStatus": "Active",
"associationSpec": {

© TM Forum 2021. All Rights Reserved.

```
      "id": "7727",
      "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/associationSpecification/7727",
      "name": "ReliesOn",
      "@referredType": "AssociationSpecfication",
      "@type": "AssociationSpecficationRef",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationSpecficationRef.schema.json"
    },
    "constraint": [
      {
        "href": "https://mycsp.com:8080/tmf-api/policyManagement/v4/policy/3947",
        "id": "3947",
        "name": "protectionDependencyRule",
        "version": "1.0",
        "@referredType": "Constraint",
        "@type": "ConstraintRef",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/ConstraintRef.schema.json"
      }
    ],
    "associationRole": [
      {
        "isSource": true,
        "role": "a_ReliesOn",
        "@type": "AssociationRole",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
        "@baseType": "",
        "entity": {
          "id": "18041",
          "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/managedEntity/18041",
          "name": "Fire Protection Electrical",
          "@type": "EntityRef",
          "@baseType": "",
          "referredType": "FireProtection",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
        }
      },
      {
        "isSource": false,
        "role": "z_ReliesOn",
        "@type": "AssociationRole",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
        "@baseType": "",
        "entity": {
          "id": "18013",
          "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/managedEntity/18013",
          "name": "Fire Protection Mechanical",
          "@type": "EntityRef",
          "@baseType": "",
          "referredType": "FireProtection",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
        }
      }
```


© TM Forum 2021. All Rights Reserved.

```
    ],
    "@type": "Association",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Association.schema.json",
    "@baseType": ""
  },
  {
    "id": "17021",
    "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/association/17021",
    "name": "HostedOn",
    "description": "A host requirement for a Tosca Database Node",
    "version": "1.0",
    "validFor": {
      "startDateTime": "2021-01-01T00:00Z",
      "endDateTime": "2023-12-31T11:59:59Z"
    },
    "lastUpdate": "2021-01-01T00:00Z",
    "lifecycleStatus": "Active",
    "associationSpec": {
      "id": "7717",
      "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/associationSpecification/7717",
      "name": "HostedOn",
      "@referredType": "AssociationSpecfication",
      "@type": "AssociationSpecficationRef",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationSpecficationRef.schema.json"
    },
    "constraint": [
      {
        "href": "https://mycsp.com:8080/tmf-api/policyManagement/v4/policy/3989",
        "id": "3989",
        "name": "EligibilityConstraint",
        "version": "1.0",
        "@referredType": "Constraint",
        "@type": "ConstraintRef",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/ConstraintRef.schema.json"
      }
    ],
    "associationRole": [
      {
        "isSource": true,
        "role": "hostee",
        "@type": "AssociationRole",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
        "@baseType": "",
        "entity": {
          "id": "18043",
          "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/managedEntity/18043",
          "name": "mysql",
          "referredType": "NodeDbms",
          "@type": "EntityRef",
          "@baseType": "",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
```


© TM Forum 2021. All Rights Reserved.

```
          }
        },
        {
          "isSource": false,
          "role": "hoster",
          "@type": "AssociationRole",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
          "@baseType": "",
          "entity": {
            "id": "18047",
            "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/managedEntity/18047",
            "name": "db_server",
            "referredType": "NodeCompute",
            "@type": "EntityRef",
            "@baseType": "",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
          }
        }
      ],
      "@type": "Association",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Association.schema.json",
      "@baseType": ""
    }
]
```


Here's an example of a request for retrieving multiple associations representing a directed graph as:

Node_1 ---> Node_2,Node_3

Node_2 ---> Node_3,Node_4

Node_3 ---> Node_4

Request

```
GET /tmf-api/entityInventory/v4/association?name=edge&group=GraphA&@type=GraphEdge
Accept: application/json
```

Response

200


```
[
  {
    "id": "17027",
    "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/association/17027",
    "name": "edge",
    "group": "GraphA",
    "description": "Represents an Uni-directional Edge of a Graph",
```


© TM Forum 2021. All Rights Reserved.

```
    "version": "1.0",
    "validFor": {
      "startDateTime": "2020-12-25T00:00Z",
      "endDateTime": "2023-12-31T11:59:59Z"
    },
    "lastUpdate": "2020-12-25T00:00Z",
    "lifecycleStatus": "Active",
    "associationSpec": {
      "id": "1901",
      "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/associationSpecification/1901",
      "name": "GraphEdgeSpec",
      "@referredType": "AssociationSpecfication",
      "@type": "AssociationSpecficationRef",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationSpecficationRef.schema.json"
    },
    "constraint": [],
    "associationRole": [
      {
        "isSource": true,
        "role": "A-Edge",
        "@type": "AssociationRole",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
        "@baseType": "",
        "entity": {
          "id": "18049",
          "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/entitySpecification/18049",
          "name": "Node-1",
          "@type": "EntityRef",
          "@baseType": "",
          "referredType": "Vertex",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
        }
      },
      {
        "isSource": false,
        "role": "Z-Edge",
        "@type": "AssociationRole",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
        "@baseType": "",
        "entity": {
          "id": "18059",
          "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/entitySpecification/18059",
          "name": "Node-2",
          "@type": "EntityRef",
          "@baseType": "",
          "referredType": "Vertex",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
        }
      }
    ],
    "@type": "GraphEdge",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/GraphEdge.schema.json",
```


© TM Forum 2021. All Rights Reserved.

```
    "@baseType": "Association"
  },
  {
    "id": "17029",
    "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/association/17029",
    "name": "edge",
    "group": "GraphA",
    "description": "Represents an Uni-directional Edge of a Graph",
    "version": "1.0",
    "validFor": {
      "startDateTime": "2020-12-25T00:00Z",
      "endDateTime": "2023-12-31T11:59:59Z"
    },
    "lastUpdate": "2020-12-25T00:00Z",
    "lifecycleStatus": "Active",
    "associationSpec": {
      "id": "1901",
      "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/associationSpecification/1901",
      "name": "GraphEdgeSpec",
      "@referredType": "AssociationSpecfication",
      "@type": "AssociationSpecficationRef",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationSpecficationRef.schema.json"
    },
    "constraint": [],
    "associationRole": [
      {
        "isSource": true,
        "role": "A-Edge",
        "@type": "AssociationRole",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
        "@baseType": "",
        "entity": {
          "id": "18049",
          "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/entitySpecification/18049",
          "name": "Node-1",
          "@type": "EntityRef",
          "@baseType": "",
          "referredType": "Vertex",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
        }
      },
      {
        "isSource": false,
        "role": "Z-Edge",
        "@type": "AssociationRole",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
        "@baseType": "",
        "entity": {
          "id": "18061",
          "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/entitySpecification/18061",
          "name": "Node-3",
          "@type": "EntityRef",
```


© TM Forum 2021. All Rights Reserved.

```
            "@baseType": "",
            "referredType": "Vertex",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
          }
        }
    ],
    "@type": "GraphEdge",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/GraphEdge.schema.json",
    "@baseType": "Association"
  },
  {
    "id": "17033",
    "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/association/17033",
    "name": "edge",
    "group": "GraphA",
    "description": "Represents an Uni-directional Edge of a Graph",
    "version": "1.0",
    "validFor": {
      "startDateTime": "2020-12-25T00:00Z",
      "endDateTime": "2023-12-31T11:59:59Z"
    },
    "lastUpdate": "2020-12-25T00:00Z",
    "lifecycleStatus": "Active",
    "associationSpec": {
      "id": "1901",
      "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/associationSpecification/1901",
      "name": "GraphEdgeSpec",
      "@referredType": "AssociationSpecfication",
      "@type": "AssociationSpecficationRef",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationSpecficationRef.schema.json"
    },
    "constraint": [],
    "associationRole": [
      {
        "isSource": true,
        "role": "A-Edge",
        "@type": "AssociationRole",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
        "@baseType": "",
        "entity": {
          "id": "18059",
          "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/entitySpecification/18059",
          "name": "Node-2",
          "@type": "EntityRef",
          "@baseType": "",
          "referredType": "Vertex",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
        }
      },
      {
        "isSource": false,
        "role": "Z-Edge",
```


© TM Forum 2021. All Rights Reserved.

```
          "@type": "AssociationRole",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
          "@baseType": "",
          "entity": {
            "id": "18077",
            "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/entitySpecification/18077",
            "name": "Node-4",
            "@type": "EntityRef",
            "@baseType": "",
            "referredType": "Vertex",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
          }
        }
    ],
    "@type": "GraphEdge",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/GraphEdge.schema.json",
    "@baseType": "Association"
  },
  {
    "id": "17041",
    "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/association/17041",
    "name": "edge",
    "group": "GraphA",
    "description": "Represents an Uni-directional Edge of a Graph",
    "version": "1.0",
    "validFor": {
      "startDateTime": "2020-12-25T00:00Z",
      "endDateTime": "2023-12-31T11:59:59Z"
    },
    "lastUpdate": "2020-12-25T00:00Z",
    "lifecycleStatus": "Active",
    "associationSpec": {
      "id": "1901",
      "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/associationSpecification/1901",
      "name": "GraphEdgeSpec",
      "@referredType": "AssociationSpecfication",
      "@type": "AssociationSpecficationRef",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationSpecficationRef.schema.json"
    },
    "constraint": [],
    "associationRole": [
      {
        "isSource": true,
        "role": "A-Edge",
        "@type": "AssociationRole",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
        "@baseType": "",
        "entity": {
          "id": "18061",
          "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/entitySpecification/18061",
          "name": "Node-3",
          "@type": "EntityRef",
```


© TM Forum 2021. All Rights Reserved.

```
            "@baseType": "",
            "referredType": "Vertex",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
          }
        },
        {
          "isSource": false,
          "role": "Z-Edge",
          "@type": "AssociationRole",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
          "@baseType": "",
          "entity": {
            "id": "18077",
            "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/entitySpecification/18077",
            "name": "Node-4",
            "@type": "EntityRef",
            "@baseType": "",
            "referredType": "Vertex",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
          }
        }
      ],
      "@type": "GraphEdge",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/GraphEdge.schema.json",
      "@baseType": "Association"
    },
    {
      "id": "17047",
      "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/association/17047",
      "name": "edge",
      "group": "GraphA",
      "description": "Represents an Uni-directional Edge of a Graph",
      "version": "1.0",
      "validFor": {
        "startDateTime": "2020-12-25T00:00Z",
        "endDateTime": "2023-12-31T11:59:59Z"
      },
      "lastUpdate": "2020-12-25T00:00Z",
      "lifecycleStatus": "Active",
      "associationSpec": {
        "id": "1901",
        "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/associationSpecification/1901",
        "name": "GraphEdgeSpec",
        "@referredType": "AssociationSpecfication",
        "@type": "AssociationSpecficationRef",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationSpecficationRef.schema.json"
      },
      "constraint": [],
      "associationRole": [
        {
          "isSource": true,
          "role": "A-Edge",
```


© TM Forum 2021. All Rights Reserved.

```
            "@type": "AssociationRole",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
            "@baseType": "",
            "entity": {
               "id": "18059",
               "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/entitySpecification/18059",
               "name": "Node-2",
               "@type": "EntityRef",
               "@baseType": "",
               "referredType": "Vertex",
               "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
            }
         },
         {
            "isSource": false,
            "role": "Z-Edge",
            "@type": "AssociationRole",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
            "@baseType": "",
            "entity": {
               "id": "18061",
               "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/entitySpecification/18061",
               "name": "Node-3",
               "@type": "EntityRef",
               "@baseType": "",
               "referredType": "Vertex",
               "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
            }
         }
      ],
      "@type": "GraphEdge",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/GraphEdge.schema.json",
      "@baseType": "Association"
   }
]
```


Retrieve association

`GET /association/{id}?fields=...&{filtering}`

Description

This operation retrieves an association entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

© TM Forum 2021. All Rights Reserved.

Usage Samples

Here's an example of a request for retrieving a specific association.

Request

```
GET /tmf-api/entityInventory/v4/association/17011
Accept: application/json
```

Response

```
200

{
   "id": "17011",
   "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/association/17011",
   "name": "ReliesOn",
   "description": "An Uni-directional ReliesOn association from Electrical Fire Protection Managed Entity to Mechanical Fire Protection Managed Entity",
   "version": "1.0",
   "validFor": {
      "startDateTime": "2020-12-25T00:00Z",
      "endDateTime": "2023-12-31T11:59:59Z"
   },
   "lastUpdate": "2020-12-25T00:00Z",
   "lifecycleStatus": "Active",
   "associationSpec": {
      "id": "7727",
      "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/associationSpecification/7727",
      "name": "ReliesOn",
      "@referredType": "AssociationSpecfication",
      "@type": "AssociationSpecficationRef",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationSpecficationRef.schema.json"
   },
   "constraint": [
      {
         "href": "https://mycsp.com:8080/tmf-api/policyManagement/v4/policy/3947",
         "id": "3947",
         "name": "protectionDependencyRule",
         "version": "1.0",
         "@referredType": "Constraint",
         "@type": "ConstraintRef",
         "@baseType": "",
         "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/ConstraintRef.schema.json"
      }
   ],
   "associationRole": [
      {
```

© TM Forum 2021. All Rights Reserved.

```
        "isSource": true,
        "role": "a_ReliesOn",
        "@type": "AssociationRole",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
        "@baseType": "",
        "entity": {
          "id": "18041",
          "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/managedEntity/18041",
          "name": "Fire Protection Electrical",
          "@type": "EntityRef",
          "@baseType": "",
          "referredType": "FireProtection",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
        }
      },
      {
        "isSource": false,
        "role": "z_ReliesOn",
        "@type": "AssociationRole",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
        "@baseType": "",
        "entity": {
          "id": "18013",
          "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/managedEntity/18013",
          "name": "Fire Protection Mechanical",
          "@type": "EntityRef",
          "@baseType": "",
          "referredType": "FireProtection",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
        }
      }
    ],
    "@type": "Association",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Association.schema.json",
    "@baseType": ""
}
```


Create association

POST /association

Description

This operation creates an association entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a Association, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

© TM Forum 2021. All Rights Reserved.

Mandatory Attributes	Rule
name	
associationRole	

Non Mandatory Attributes	Rule
@baseType	
@schemaLocation	
@type	
associationSpec	
constraint	
description	
lastUpdate	
lifecycleStatus	
validFor	
version	

Usage Samples

Here's an example of a request for creating an association.

Request

```
POST /tmf-api/entityInventory/v4/association
Content-Type: application/json

{
   "name": "ReliesOn",
   "description": "An Uni-directional ReliesOn association from Electrical Fire Protection Managed Entity to Mechanical Fire Protection Managed Entity",
   "version": "1.0",
   "validFor": {
      "startDateTime": "2020-12-25T00:00Z",
      "endDateTime": "2023-12-31T11:59:59Z"
   },
   "lastUpdate": "2020-12-25T00:00Z",
   "lifecycleStatus": "Active",
   "associationSpec": {
      "id": "7727",
      "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/associationSpecification/7727",
      "name": "ReliesOn",
      "@referredType": "AssociationSpecfication",
      "@type": "AssociationSpecficationRef",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationSpecficationRef.schema.json"
   },
   "constraint": [
      {
         "href": "https://mycsp.com:8080/tmf-api/policyManagement/v4/policy/3947",
         "id": "3947",
```

© TM Forum 2021. All Rights Reserved.

```
        "name": "protectionDependencyRule",
        "version": "1.0",
        "@referredType": "Constraint",
        "@type": "ConstraintRef",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/ConstraintRef.schema.json"
      }
    ],
    "associationRole": [
      {
        "isSource": true,
        "role": "a_ReliesOn",
        "@type": "AssociationRole",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
        "entity": {
          "id": "18041",
          "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/managedEntity/18041",
          "name": "Fire Protection Electrical",
          "@type": "EntityRef",
          "referredType": "FireProtection",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
        }
      },
      {
        "isSource": false,
        "role": "z_ReliesOn",
        "@type": "AssociationRole",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
        "entity": {
          "id": "18013",
          "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/managedEntity/18013",
          "name": "Fire Protection Mechanical",
          "@type": "EntityRef",
          "referredType": "FireProtection",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
        }
      }
    ],
    "@type": "Association",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Association.schema.json"
}
```


Response

201


```
{
    "id": "17011",
    "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/association/17011",
    "name": "ReliesOn",
    "description": "An Uni-directional ReliesOn association from Electrical Fire Protection Managed Entity to Mechanical Fire Protection Managed Entity",
```


© TM Forum 2021. All Rights Reserved.

```
  "version": "1.0",
  "validFor": {
    "startDateTime": "2020-12-25T00:00Z",
    "endDateTime": "2023-12-31T11:59:59Z"
  },
  "lastUpdate": "2020-12-25T00:00Z",
  "lifecycleStatus": "Active",
  "associationSpec": {
    "id": "7727",
    "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/associationSpecification/7727",
    "name": "ReliesOn",
    "@referredType": "AssociationSpecfication",
    "@type": "AssociationSpecficationRef",
    "@baseType": "",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationSpecficationRef.schema.json"
  },
  "constraint": [
    {
      "href": "https://mycsp.com:8080/tmf-api/policyManagement/v4/policy/3947",
      "id": "3947",
      "name": "protectionDependencyRule",
      "version": "1.0",
      "@referredType": "Constraint",
      "@type": "ConstraintRef",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/ConstraintRef.schema.json"
    }
  ],
  "associationRole": [
    {
      "isSource": true,
      "role": "a_ReliesOn",
      "@type": "AssociationRole",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
      "@baseType": "",
      "entity": {
        "id": "18041",
        "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/managedEntity/18041",
        "name": "Fire Protection Electrical",
        "@type": "EntityRef",
        "@baseType": "",
        "referredType": "FireProtection",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
      }
    },
    {
      "isSource": false,
      "role": "z_ReliesOn",
      "@type": "AssociationRole",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
      "@baseType": "",
      "entity": {
        "id": "18013",
```


© TM Forum 2021. All Rights Reserved.

```
            "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/managedEntity/18013",
            "name": "Fire Protection Mechanical",
            "@type": "EntityRef",
            "@baseType": "",
            "referredType": "FireProtection",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
          }
        }
    ],
    "@type": "Association",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Association.schema.json",
    "@baseType": ""
}
```


Patch association

PATCH /association/{id}

Description

This operation allows partial updates of an association entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional.

Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

The tables below provide the list of patchable and non patchable attributes, including constraint rules on their usage.

Patchable Attributes	Rule
associationRole	
associationSpec	
constraint	
description	
lifecycleStatus	
name	
validFor	
version	

Non Patchable Attributes	Rule
id	
href	
@type	

© TM Forum 2021. All Rights Reserved.

Non Patchable Attributes	Rule
@baseType	
lastUpdate	
@schemaLocation	

Usage Samples

Here's an example of a request for patching a Association resource.

Request

```
PATCH /tmf-api/entityInventory/v4/association/17011
Content-Type: application/merge-patch+json

{
   "name": "new name"
}
```

Response

```
200

{
   "id": "17011",
   "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/association/17011",
   "name": "new name",
   "description": "An Uni-directional ReliesOn association from Electrical Fire Protection Managed Entity to Mechanical Fire Protection Managed Entity",
   "version": "1.0",
   "validFor": {
      "startDateTime": "2020-12-25T00:00Z",
      "endDateTime": "2023-12-31T11:59:59Z"
   },
   "lastUpdate": "2020-12-25T00:00Z",
   "lifecycleStatus": "Active",
   "associationSpec": {
      "id": "7727",
      "href": "https://mycsp.com:8080/tmf-api/entityCatalogManagement/v4/associationSpecification/7727",
      "name": "ReliesOn",
      "@referredType": "AssociationSpecfication",
      "@type": "AssociationSpecficationRef",
      "@baseType": "",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationSpecficationRef.schema.json"
   },
   "constraint": [
      {
         "href": "https://mycsp.com:8080/tmf-api/policyManagement/v4/policy/3947",
         "id": "3947",
```

© TM Forum 2021. All Rights Reserved.

```
        "name": "protectionDependencyRule",
        "version": "1.0",
        "@referredType": "Constraint",
        "@type": "ConstraintRef",
        "@baseType": "",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/ConstraintRef.schema.json"
      }
    ],
    "associationRole": [
      {
        "isSource": true,
        "role": "a_ReliesOn",
        "@type": "AssociationRole",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
        "@baseType": "",
        "entity": {
          "id": "18041",
          "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/managedEntity/18041",
          "name": "Fire Protection Electrical",
          "@type": "EntityRef",
          "@baseType": "",
          "referredType": "FireProtection",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
        }
      },
      {
        "isSource": false,
        "role": "z_ReliesOn",
        "@type": "AssociationRole",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/AssociationRole.schema.json",
        "@baseType": "",
        "entity": {
          "id": "18013",
          "href": "https://mycsp.com:8080/tmf-api/entityInventory/v4/managedEntity/18013",
          "name": "Fire Protection Mechanical",
          "@type": "EntityRef",
          "@baseType": "",
          "referredType": "FireProtection",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/EntityRef.schema.json"
        }
      }
    ],
    "@type": "Association",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Common/Association.schema.json",
    "@baseType": ""
}
```


© TM Forum 2021. All Rights Reserved.

Delete association

`DELETE /association/{id}`

Description

This operation deletes an association entity.

Usage Samples

Here's an example of a request for deleting an association.

Request
DELETE /tmf-api/entityInventory/v4/association/17011
Response
204

© TM Forum 2021. All Rights Reserved.

API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

`POST /hub`

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request
POST /api/hub Accept: application/json {"callback": "http://in.listener.com"}
Response
201 Content-Type: application/json Location: /api/hub/42 {"id":"42","callback":"http://in.listener.com","query":null}

© TM Forum 2021. All Rights Reserved.

Unregister listener

`DELETE /hub/{id}`

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request
DELETE /api/hub/42 Accept: application/json
Response
204

Publish Event to listener

`POST /client/listener`

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

© TM Forum 2021. All Rights Reserved.

Usage Samples

Here's an example of a notification received by the listener. In this example “EVENT TYPE” should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

POST /client/listener
Accept: application/json
 	
{
"event": {
EVENT BODY
},
"eventType": "EVENT_TYPE"
}

Response
201

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

© TM Forum 2021. All Rights Reserved.

ROLE BASED ACCESS CONTROL

The API will be released to allow all CRUD operations on API REST resources. However, the API operations are in subject to access control and can be limited depends on API user role. After a user presents his credentials for authentication, depending his validity of credentials he is given access to catalog based on his role.

For example, a User gets an error response if the user issues a Get on REST resources and he has no permission to access to those resources or if he tries to manipulate a resource but do not have Write Access.

Normally we anticipate that the OAUTH2 or Open ID Connect are used as the authorization APIs and that ACL are established between authorized parties with regards to the content of the inventory (i.e., GET but also enable of update operations on specific entities).

© TM Forum 2021. All Rights Reserved.

Acknowledgements

Document History

Version History

Version Number	Date	Release led by:	Description
Release 1.0	15-Apr-2017	Pierre Gauthier TM Forum Mariano Belaunde, Orange Labs	First Release of the Document.
Release 2.0	06-Nov-2018	Mariano Belaunde, Orange Labs	Alignment with Guidelines 3.0
Release 3.0	08-Apr-2021	Lara Silva, Vodafone lara.silva3@vodafone.com	Alignment with guidelines 4.0.2
4.0.0	15-Sep-2021	Kamal Maghsoudlou	

Release History

Release Status	Date	Release led by:	Description
18.5.1	05-Apr-2019		TM Forum Approved
Pre-Production	15-Sep-2011	Kamal Maghsoudlou	
Production	19-Nov-2021	Adrienne Walcott	Updated to reflect TM Forum Approved Status

© TM Forum 2021. All Rights Reserved.

CONTRIBUTORS TO DOCUMENT

- Kamal Maghsoudlou, Ericsson kamal.maghsoudlou@ericsson.com
- Pierre Gauthier, TM Forum pgauthier@tmforum.org

© TM Forum 2021. All Rights Reserved.